



OPEN ACCESS

EDITED BY

Soren Bohos Hayrabedyan,
Institute of Biology and Immunology of
Reproduction (BAS), Bulgaria

REVIEWED BY

Julio C. Bueno,
University of Antioquia, Colombia
Le Bo,
The First Affiliated Hospital of Soochow
University, China

*CORRESPONDENCE

Xiaoling Feng
✉ doctorwsm@163.com

RECEIVED 19 November 2025

REVISED 02 July 2026

ACCEPTED 06 August 2026

PUBLISHED 28 August 2026

CITATION

Wang S, Guo Q, Niu D, Chen Y, Ping Y,
Ren J, Qi D and Feng X (2026) The β -
hCG-TNF- α -T cell axis in threatened
abortion: unraveling the pathogenic
cascade from aberrant glycosylation to
inflammatory demise.
Front. Immunol. 17:1749672.
doi: 10.3389/fimmu.2026.1749672

COPYRIGHT

© 2026 Wang, Guo, Niu, Chen, Ping, Ren,
Qi and Feng. This is an open-access
article distributed under the terms of the
[Creative Commons Attribution License](#)
(CC BY). The use, distribution or
reproduction in other forums is
permitted, provided the original
author(s) and the copyright owner(s) are
credited and that the original publication
in this journal is cited, in accordance
with accepted academic practice. No
use, distribution or reproduction is
permitted which does not comply with
these terms.

The β -hCG-TNF- α -T cell axis in threatened abortion: unraveling the pathogenic cascade from aberrant glycosylation to inflammatory demise

Shimeng Wang¹, Qing Guo², Dingren Niu³, Yao Chen³,
Yunlu Ping¹, Jiajie Ren³, Ding Qi⁴ and Xiaoling Feng^{1*}

¹The 1st Affiliated Hospital of Heilongjiang University of Traditional Chinese Medicine, Harbin, Heilongjiang, China, ²Heilongjiang Academy of Traditional Chinese Medicine, Harbin, Heilongjiang, China, ³Heilongjiang University of Chinese Medicine, Harbin, Heilongjiang, China, ⁴The 2nd Affiliated Hospital of Heilongjiang University of Traditional Chinese Medicine, Harbin, Heilongjiang, China

Threatened abortion (TA), vaginal bleeding before 20 weeks in a viable pregnancy, affects 20-25% of early pregnancies. Its unpredictable nature is a major clinical challenge. Current diagnostics (β -hCG and ultrasound) are retrospective and fail to reveal the dynamic pathology. This review presents the "Danger Signal" Hypothesis, focusing on the β -hCG-TNF- α -T cell axis, to explain TA pathogenesis. The process involves three stages: 1) Initiation: Stressed trophoblasts secrete aberrantly glycosylated β -hCG, which acts as a DAMP. 2) Amplification: This DAMP triggers a TNF- α -dominated cytokine storm, reprogramming T cell metabolism to glycolysis and sustaining inflammation. 3) Execution: The inflammatory microenvironment induces trophoblast pyroptosis, creating a vicious cycle of damage. We synthesize evidence from glycobiology, immunometabolism, and cell death research to substantiate this model, explore its implications for the development of stage-specific biomarkers (such as β -hCG glycoforms and GSDMD-N fragments), and discuss precision therapeutic approaches (including the use of metformin for metabolic intervention and anti-TNF- α agents for managing late-phase inflammation). This mechanistic framework aims to guide future research and improve clinical outcomes.

KEYWORDS

danger signal hypothesis, immunometabolism, pyroptosis, threatened abortion, TNF- α , β -hCG glycosylation

1 Introduction

1.1 Clinical burden and unmet need of threatened abortion

Threatened abortion (TA) is the most common complication in early pregnancy, affecting approximately 20-25% of all gestations (1, 2). Clinically defined by vaginal bleeding (with or without abdominal cramping) before 20 weeks of gestation in the presence of a viable intrauterine fetus, TA presents a significant diagnostic and therapeutic dilemma:

while 75-85% of cases progress to uneventful pregnancy continuation, 15-25% culminate in irreversible miscarriage (3). This prognostic uncertainty not only induces considerable psychological distress in pregnant individuals but also contributes to suboptimal clinical management. This mismanagement ranges from the excessive use of empirical interventions, such as progesterone supplementation, to delayed recognition of progressive pregnancy loss (4).

Current clinical monitoring strategies primarily involve serial transvaginal ultrasound to evaluate fetal viability and placental location, alongside quantitative measurement of serum biomarkers, particularly β -hCG and progesterone (5). However, these approaches have inherent limitations: declining β -hCG levels or low progesterone concentrations reflect established trophoblast dysfunction rather than predicting the onset of pathological processes. Notably, serum β -hCG quantification fails to distinguish between transient trophoblast stress (reversible) and irreversible cellular damage, highlighting the critical need for mechanistic insights into TA pathogenesis to identify predictive biomarkers and targeted therapies.

1.2 Etiological hypotheses and the need for a unified model

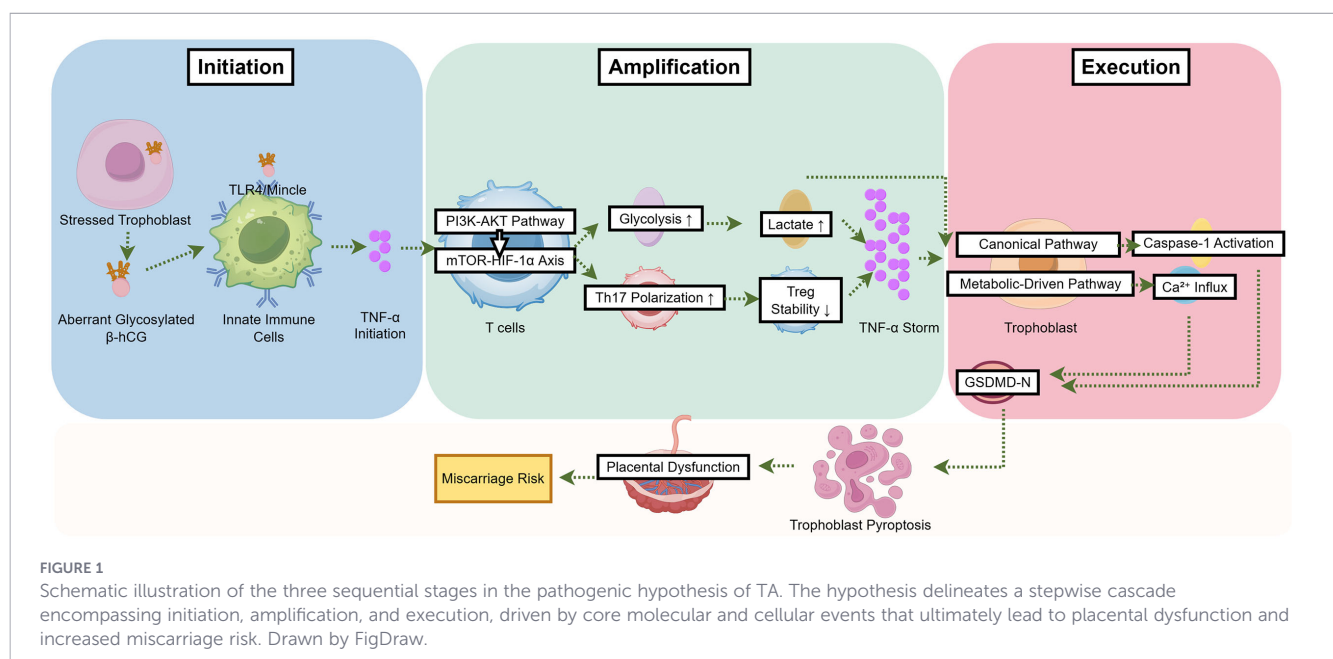
The etiology of TA is widely recognized as multifactorial, with several key contributing factors extensively documented in the literature. These include endocrine disorders, such as luteal phase defect and thyroid dysfunction (6), impairments in maternal immune tolerance, exemplified by Th17/Treg cell imbalance and aberrant cytokine profiles (7, 8), thrombophilic conditions, notably antiphospholipid syndrome (9), and environmental stressors, including oxidative stress and nutrient deprivation (10). Although these studies have identified associations between specific phenotypic changes and TA, they predominantly remain descriptive and

do not elucidate the dynamic sequence of molecular events that facilitate the transition from a reversible “at-risk” state to irreversible pregnancy loss.

A significant knowledge gap persists in understanding the upstream triggers of inflammation and the downstream effectors that sustain a maladaptive immune response. For instance, elevated levels of tumor necrosis factor- α (TNF- α) and skewed Th17/Treg ratios are consistently observed in TA (11, 12), yet the mechanisms by which these immune perturbations are initiated and maintained remain unclear. This lack of mechanistic understanding has hindered the development of targeted therapies, with current interventions limited to non-specific symptom management rather than addressing the root cause of TA.

1.3 The “danger signal” hypothesis: a novel framework for TA pathogenesis

To address the existing gap in understanding TA pathogenesis, we propose the “Danger Signal” Hypothesis—a comprehensive and dynamic model that integrates glycobiology, immunometabolism, and cell death pathways to elucidate the progression of TA. This hypothesis suggests that TA is initiated by qualitative alterations in trophoblast-derived signals, as opposed to quantitative changes, which are misinterpreted by the maternal immune system as “danger” cues. This misinterpretation triggers a self-perpetuating cycle of inflammation and tissue damage. The hypothesis delineates three sequential stages (Figure 1): initiation through aberrant β -hCG glycosylation, amplification via TNF- α -mediated metabolic reprogramming, and execution by trophoblast pyroptosis. This review consolidates recent evidence to substantiate each stage of the hypothesis, examines its translational implications for the development of biomarkers and therapeutic interventions, and outlines future research directions to enhance TA management.



2 The initiation stage: aberrantly glycosylated β -hCG as a decoy danger signal

2.1 β -hCG glycosylation: a key regulator of its tolerogenic function

Human chorionic gonadotropin (hCG) is a heterodimeric glycoprotein consisting of an α -subunit, which is shared with luteinizing hormone, follicle-stimulating hormone, and thyroid-stimulating hormone, and a distinct β -subunit (β -hCG) (13). The β -subunit is extensively glycosylated, possessing four N-linked glycosylation sites (Asn13, Asn30, Asn63, Asn74) and one O-linked site (Ser121) (14). The composition and structural features of these glycan chains, including branching, sialylation, and fucosylation, are not merely structural modifications but are essential determinants of the biological activity and stability of β -hCG (15).

During normal early pregnancy, trophoblasts secrete β -hCG with a specific glycosylation profile that facilitates maternal immune tolerance towards the semi-allogeneic fetus. Proper glycosylation of β -hCG induces tolerogenic effects through several mechanisms (Figure 2): (1) it promotes the expansion of regulatory T cells (Tregs) via activation of the TGF- β /Smad signaling pathway (16);

(2) it polarizes decidual macrophages towards the anti-inflammatory M2 phenotype by upregulating IL-10 and downregulating TNF- α (17); (3) it suppresses the cytotoxicity of natural killer (NK) cells by inhibiting the expression of activating receptors, such as NKG2D (17, 18); (4) enhancing angiogenesis at the maternal-fetal interface via activation of the VEGF pathway (19). Collectively, these functions establish β -hCG as a central mediator of pregnancy maintenance, with its glycosylation profile serving as a “molecular signature” of trophoblast health.

2.2 Stress-induced alterations in β -hCG glycosylation and DAMP transformation

Trophoblasts exhibit a high susceptibility to cellular stressors, which are prevalent during early pregnancy and become exacerbated under pathological conditions. Principal stressors include oxidative stress, arising from inadequate placental perfusion; endoplasmic reticulum (ER) stress, resulting from impaired protein folding; and nutrient deprivation, such as deficiencies in glucose or amino acids (20, 21). These stressors disrupt the enzymatic machinery involved in glycosylation, including glycosyltransferases (e.g., GnT-IV, GnT-V) and glycosidases, leading to the production of β -hCG with immature or aberrant glycan structures (22, 23).

A well-characterized aberrant glycoform is hyperglycosylated hCG (hCG-h). During normal pregnancy, hCG-h is the predominant

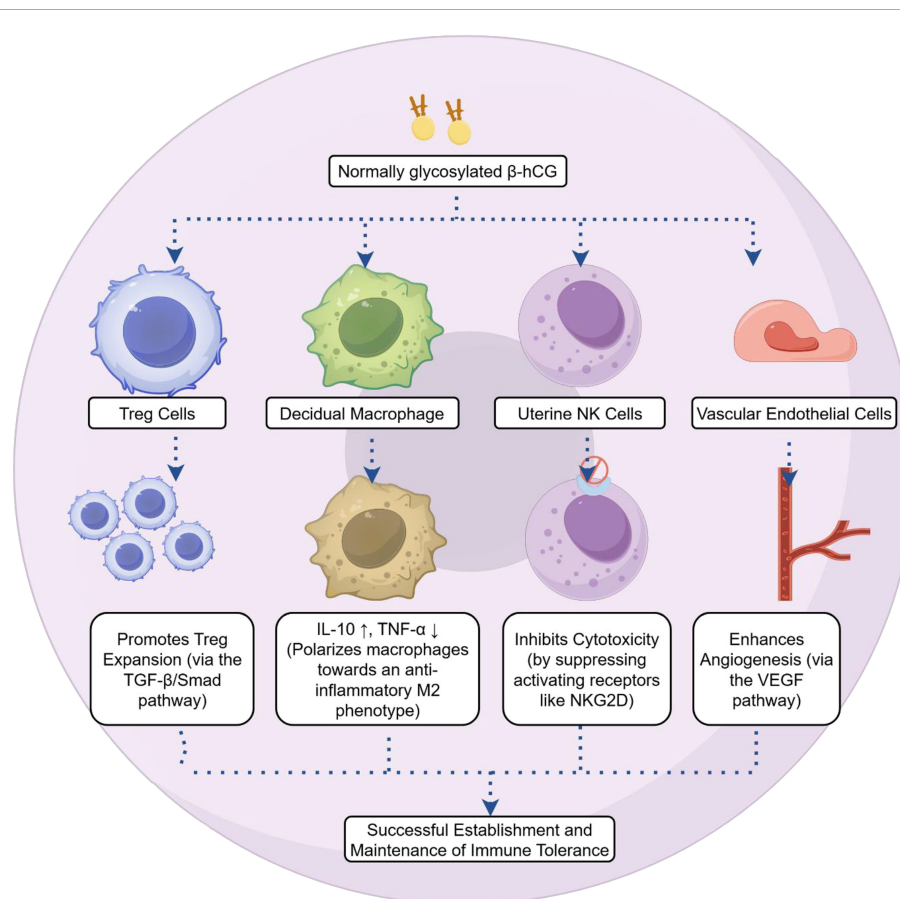


FIGURE 2

Mechanistic diagram illustrating the role of β -hCG with specific glycosylation features secreted by normal trophoblasts in establishing immune tolerance and promoting angiogenesis during pregnancy. Drawn by FigDraw.

form secreted by invasive cytotrophoblasts in the first 6–8 weeks of gestation, where it plays a critical role in placental implantation (24). Prospective studies have consistently shown that higher hCG-h/total hCG ratios at implantation are associated with ongoing pregnancies (25). Beyond this window, the syncytiotrophoblast becomes the dominant hCG source, and glycosylation normally shifts toward less branched, fully sialylated isoforms. Persistent hCG-h production by syncytiotrophoblasts thereafter represents a maladaptive reversion to a primitive glycosylation pattern, indicative of trophoblast stress, and has been linked to early miscarriage (26, 27).

The structural anomalies of hCG-h are defined by two key alterations (28). First, there is a significant extension and complexification of O-linked glycans on the β -subunit's C-terminal peptide (CTP), where larger 6-sugar structures become predominant, contrasting sharply with the simpler cores of native hCG. Second, the N-linked glycans exhibit a marked increase in tri-antennary (branched) structures, a change functionally analogous to the upregulation of GnT-V. Concurrently, a significant loss of terminal sialic acid residues exposes underlying sugar motifs, such as galactose and N-acetylglucosamine (GlcNAc), which are typically “capped” and hidden. These changes are functionally consistent with upregulated GnT-V activity and impaired sialyltransferase function under stress conditions (23).

These structural shifts fundamentally alter receptor engagement. The highly branched N-glycans structurally resemble known TLR4 ligands such as lipopolysaccharide (29), potentially promoting aberrant TLR4/NF- κ B activation (30). Concurrently, exposed

terminal sugars (Gal, GlcNAc) resemble ligands for C-type lectin receptors such as Mincle (31), while enhanced core fucosylation via FUT8 may also engage lectin receptors (32). Although direct biophysical evidence of β -hCG binding to TLR4/Mincle remains limited, the striking structural homology to pathogen-associated molecular patterns (PAMPs) strongly implies promiscuous engagement with these PRRs (33). Consequently, the glycan profile shifts from tolerogenic lectins (Galectin-1, DC-SIGN) toward inflammatory PRRs (TLR4, Mincle), initiating an inflammatory cascade.

While existing *in vitro* models, such as hypoxia-reoxygenation in HTR-8/SVneo cells, have demonstrated that oxidative stress can enhance glycosyltransferase activity, including GnT-V, thereby suggesting an influence on β -hCG glycosylation, there is still an absence of direct evidence connecting oxidative stress, such as H₂O₂ treatment, in BeWo cells or primary trophoblasts to increased hCG-h secretion or reduced sialyltransferase activity (34, 35). Establishing this causal link between stress and specific glycosylation phenotypes remains a key step in confirming the upstream trigger of the “Danger Signal” hypothesis.

As summarized in Figure 3, we hypothesize that stress-induced aberrant glycosylation converts β -hCG from a tolerogenic signal into a DAMP. Aberrant β -hCG glycoforms engage TLR4 on decidual macrophages, activating NF- κ B, and Mincle on DCs, triggering inflammasome activation. This inappropriate PRR engagement causes the maternal immune system to erroneously recognize aberrant β -hCG as a danger signal, initiating a pathological cascade.

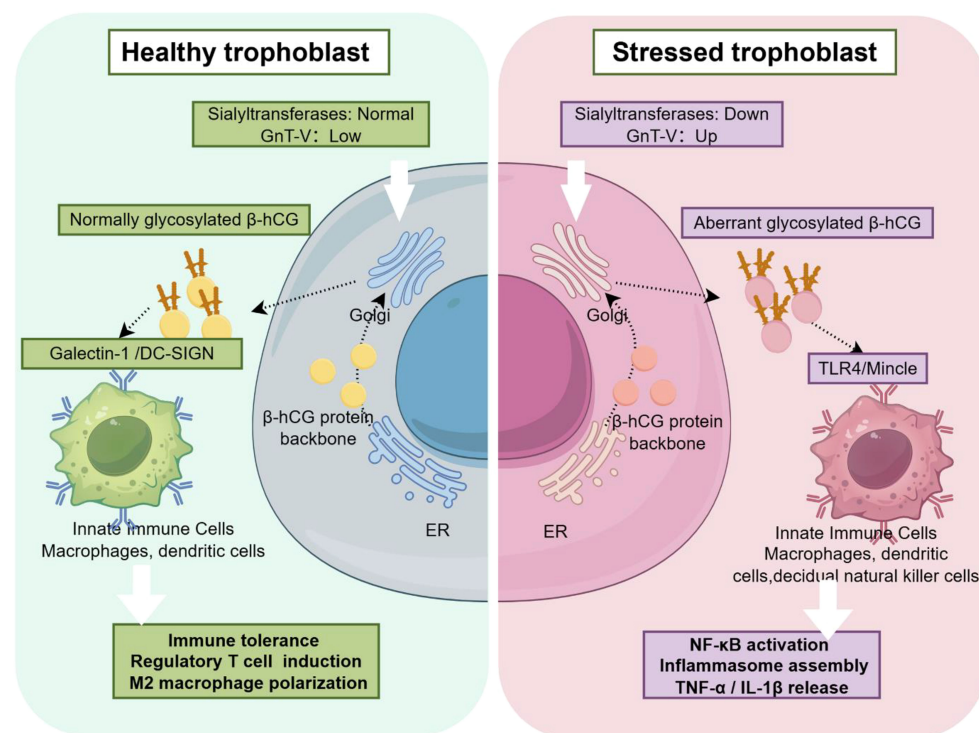


FIGURE 3

Glycosylation-mediated immune modulation by β -hCG at the fetomaternal interface. Healthy trophoblasts secrete normally glycosylated β -hCG that binds tolerogenic lectins (Galectin-1/DC-SIGN) to maintain immune tolerance. Stressed trophoblasts produce aberrantly glycosylated β -hCG, which fails to engage tolerogenic receptors but instead activates pro-inflammatory PRRs (TLR4/Mincle). This aberrant glycoform drives NF- κ B signaling and inflammasome assembly in innate immune cells (macrophages, DCs, dNK cells), initiating a pathological inflammatory cascade. ER: Endoplasmic Reticulum; Golgi: Golgi apparatus. Drawn by FigDraw.

3 The amplification stage: TNF- α storm and T cell metabolic reprogramming

3.1 TNF- α : the central amplifier of inflammation

The recognition of aberrant β -hCG by innate immune cells induces the secretion of pro-inflammatory cytokines, with TNF- α playing a pivotal role in the amplification phase. In addition to decidual macrophages and dendritic cells (DCs), the abundant decidual natural killer (dNK) cells at the implantation site may directly sense the aberrant glycoforms of β -hCG via surface PRRs (36). We hypothesize that this interaction could potentially trigger dNK cells degranulation and the secretion of TNF- α and IFN- γ , contributing to the initial propagation of the inflammatory cascade. Consequently, TNF- α levels in both serum and decidual tissue are markedly elevated in women with threatened abortion (TA) who subsequently experience miscarriage (37, 38). A meta-analysis comprising 11 studies ($n=1371$) revealed that the standardized mean difference in TNF- α levels was 2.82 units (95% confidence interval: 1.57-4.06) in patients who experienced miscarriage compared to those with ongoing pregnancies ($P < 0.0001$) (39).

TNF- α perpetuates an inflammatory feedback loop through two principal mechanisms, as illustrated in Figure 4: (1) Autocrine activation, wherein TNF- α binds to its receptors (TNFR1/2) on innate immune cells, subsequently activating the NF- κ B pathway, which enhances the transcription of TNF- α (40, 41); (2) Paracrine recruitment, whereby TNF- α stimulates the expression of chemokines, such

as CXCL8 and CCL2, which attract additional pro-inflammatory cells, including neutrophils and Th17 cells, to the maternal-fetal interface (42, 43). This process culminates in a “TNF- α storm,” characterized by a nonlinear escalation in local TNF- α concentrations that surpasses the capacity of endogenous anti-inflammatory mechanisms, such as soluble TNFRs and IL-10, to mitigate (44, 45).

Importantly, the shift from physiological TNF- α , which is integral to normal placental remodeling, to a pathological TNF- α storm is delineated by a critical threshold. Although a universally accepted absolute concentration threshold for TNF- α in early pregnancy remains unestablished—largely due to inter-individual variability, methodological differences across assays, and the transient nature of cytokine kinetics, a functional threshold can be conceptually defined. Rather than relying on absolute serum levels, this threshold is better characterized by the balance between pro-inflammatory and regulatory mediators. Specifically, a sustained shift in the TNF- α /IL-10 ratio or an elevated free TNF- α to soluble receptor (sTNFR2) ratio reflects a transition from physiological immune adaptation to a pathological inflammatory state (12, 39). This functional tipping point corresponds to the “point of no return,” wherein transient NF- κ B signaling gives way to sustained activation, driving trophoblast apoptosis and establishing a self-perpetuating inflammatory loop (46). Definitive quantitative cutoffs for clinical application await validation in large, standardized prospective cohorts.

3.2 Immunometabolic reprogramming: the mechanism of irreversible inflammation

The “Danger Signal” Hypothesis presents a novel perspective by identifying T cell metabolic reprogramming as the mechanism through

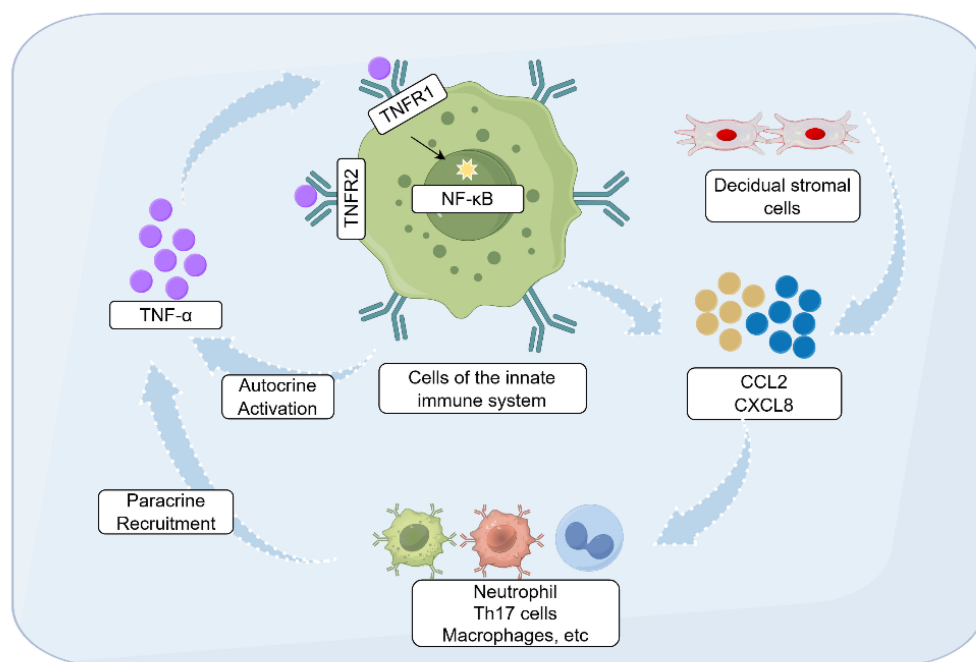


FIGURE 4

Schematic illustration of the two key mechanisms by which TNF- α drives a self-perpetuating inflammatory loop. TNF- α can bind to TNFR1 and TNFR2 on the surface of immune cells to activate NF- κ B signaling pathway, achieving autocrine activation of innate immune cells. Meanwhile, activated innate immune cells secrete chemokines such as CCL2 and CXCL8 to recruit immune cells including neutrophils, Th17 cells, and macrophages from the periphery (paracrine recruitment), participating in the initiation and regulation of immune responses. Drawn by FigDraw.

which $\text{TNF-}\alpha$ maintains the immune system in a pro-inflammatory state. The functionality of immune cells is intricately linked to their metabolic pathways: pro-inflammatory T cells, such as Th1 and Th17, predominantly utilize aerobic glycolysis (known as the Warburg effect) to facilitate rapid energy production and biomass synthesis. In contrast, Tregs rely on oxidative phosphorylation (OXPHOS) and fatty acid oxidation (FAO) to exert their suppressive functions (47, 48).

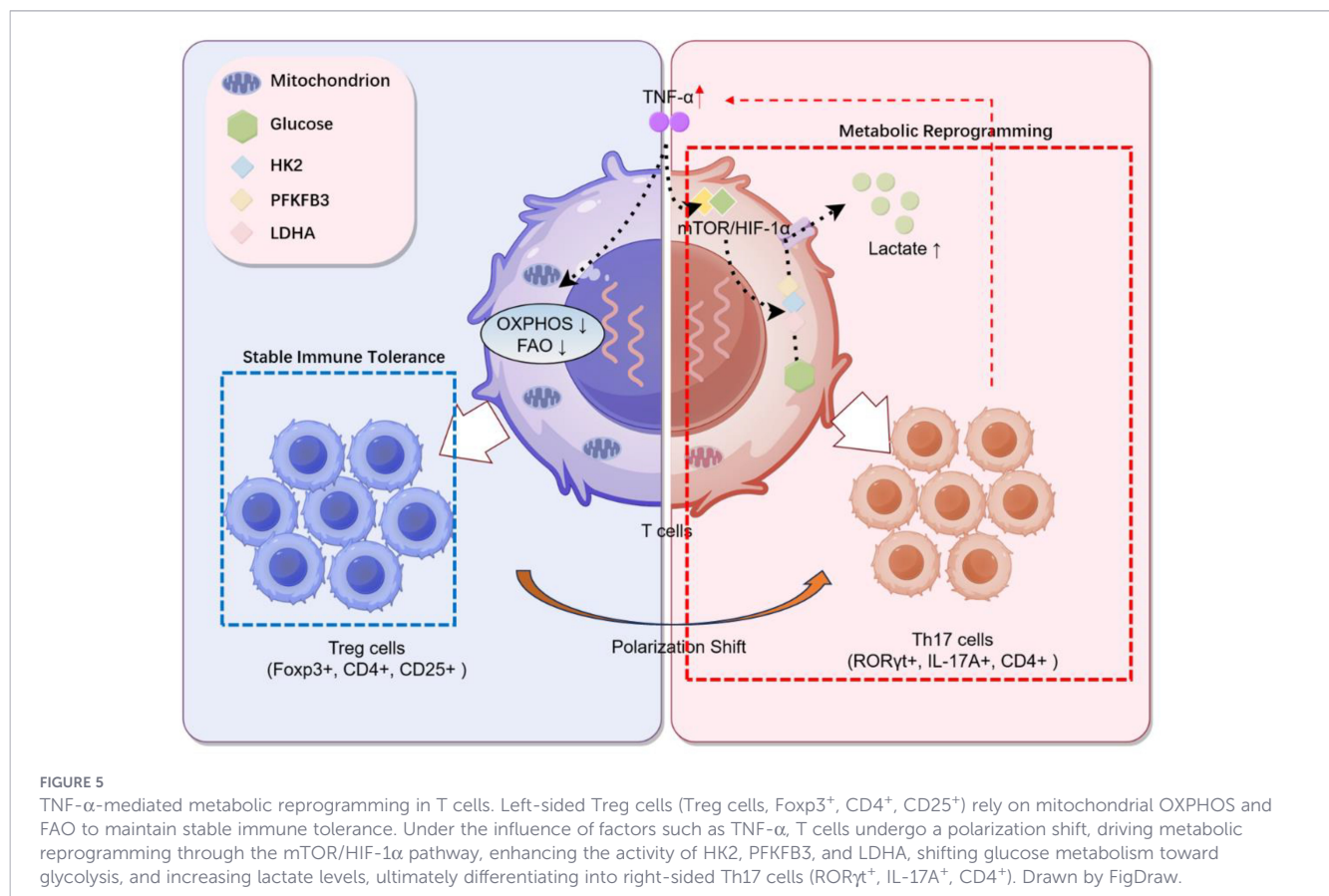
$\text{TNF-}\alpha$ serves as a pivotal regulator of this metabolic equilibrium by promoting a shift in T cell metabolism towards glycolysis via activation of the mTOR-HIF-1 α axis (Figure 5). Initially, the binding of $\text{TNF-}\alpha$ to TNFR1 triggers the PI3K-AKT signaling pathway, leading to the phosphorylation and activation of mTOR, a critical metabolic sensor that enhances glycolysis by upregulating the expression of glucose transporters (e.g., GLUT1) and glycolytic enzymes (e.g., hexokinase 2, lactate dehydrogenase A) (49, 50). Secondly, the activation of the mechanistic target of rapamycin (mTOR), in conjunction with $\text{TNF-}\alpha$ -induced production of reactive oxygen species (ROS), leads to the stabilization of hypoxia-inducible factor-1 alpha (HIF-1 α) (51). HIF-1 α is a transcription factor that promotes the expression of glycolytic genes, such as 6-phosphofructo-2-kinase/fructose-2, 6-bisphosphatase 3 (PFKFB3), and transcription factors specific to Th17 cells, such as retinoic acid-related orphan receptor gamma t (ROR γ t) (52). Simultaneously, $\text{TNF-}\alpha$ compromises mitochondrial function by inducing ROS-mediated damage to mitochondrial DNA (mtDNA) and inhibiting the activity of complex I within the electron transport chain (53, 54). This results in a reduction of oxidative phosphorylation (OXPHOS) and fatty acid oxidation (FAO), thereby undermining the stability and function of regulatory T cells (Tregs).

The “point of no return” in the progression of TA is characterized by the completion of this metabolic reprogramming; once T cells transition from OXPHOS to glycolysis, the pro-inflammatory phenotype becomes self-perpetuating through metabolic-transcriptional and metabolic-cytokine feedback mechanisms.

4 Trophoblast pyroptosis: the execution stage of inflammatory demise

4.1 Mechanisms of trophoblast pyroptosis in TA

While apoptosis (programmed, non-inflammatory cell death) has long been recognized in trophoblast dysfunction (55), recent evidence identifies pyroptosis as the dominant cell death mechanism driving irreversible tissue damage in TA. Pyroptosis is mediated by gasdermin (GSDM) family proteins, particularly gasdermin D (GSDMD), which form pores in the plasma membrane upon cleavage by inflammatory caspases (caspase-1/4/5/11) (56). The formation of pores in the cellular membrane results in cellular swelling, the release of intracellular components—including damage-associated molecular patterns (DAMPs) and pro-inflammatory cytokines—and the subsequent amplification of the inflammatory response, which is markedly different from the silent clearance of apoptotic cells mediated by phagocytosis (57).



In the context of TA, trophoblast pyroptosis is triggered by two converging pathways: the canonical inflammasome pathway and the metabolically driven pathway (Figure 6). The canonical pathway is activated by DAMPs released from stressed trophoblasts, such as aberrant β -hCG and mtDNA, as well as by pathogens, including those responsible for intrauterine infections. These stimuli activate nucleotide-binding oligomerization domain (NOD)-like receptors (NLRs), including NLRP3, which assemble into inflammasomes—multiprotein complexes that facilitate the recruitment and activation of caspase-1 (58). Once activated, caspase-1 cleaves pro-IL-1 β and pro-IL-18 into their mature forms and processes GSDMD into its active N-terminal fragment (GSDMD-N), thereby initiating pyroptosis (59).

The metabolically driven pathway, which is specific to TA, is triggered by the glycolytic microenvironment established during the amplification phase. Elevated lactate levels (a byproduct of T cell glycolysis) reduce intracellular pH, activating acid-sensing ion channels (ASICs) on trophoblasts (60). ASIC activation induces calcium influx, which activates calpain proteases—enzymes that directly cleave GSDMD into GSDMD-N, thus providing a non-canonical, caspase-independent route to pyroptosis. This calpain-mediated pathway operates in parallel with the canonical inflammasome pathway, explaining why pyroptosis persists even in the absence of canonical inflammasome activation (61). This caspase-independent pathway explains why pyroptosis persists even in the absence of canonical inflammasome activation.

4.2 Pyroptosis-mediated vicious cycle of inflammation

The execution stage completes the pathogenic cascade by creating a vicious cycle of inflammation and tissue damage (Figure 7). Pyroptotic trophoblasts release three classes of pro-inflammatory mediators that perpetuate the pathological response: DAMPs, mature cytokines, and aberrant glycan fragments.

First, DAMPs such as mtDNA, high-mobility group box 1 (HMGB1), and heat shock proteins (HSPs) are released into the decidual microenvironment upon trophoblast pyroptosis (62). These DAMPs interact with PRRs on innate immune cells; for instance, mtDNA activates Toll-like receptor 9 (TLR9), while HMGB1 engages Toll-like receptor 4 (TLR4), thereby further stimulating the secretion of TNF- α and IL-1 β (63). A study by Huang et al. (64) demonstrated that HMGB1 released from pyroptotic trophoblasts elevates the production of IL-18, IL-1 β , and TNF- α by decidual macrophages, thereby reinforcing the cytokine storm characteristic of the amplification stage.

Furthermore, mature pro-inflammatory cytokines (IL-1 β and IL-18), cleaved during pyroptosis, enhance the metabolic skewing and pro-inflammatory polarization of T cells. IL-1 β activates the mTOR- HIF-1 α axis in T cells, leading to increased expression of glucose transporter 1 (GLUT1) and lactate production (65). Concurrently, IL-18 promotes interferon-gamma (IFN- γ) secretion by Th1 cells, which suppresses Treg function and further disrupts

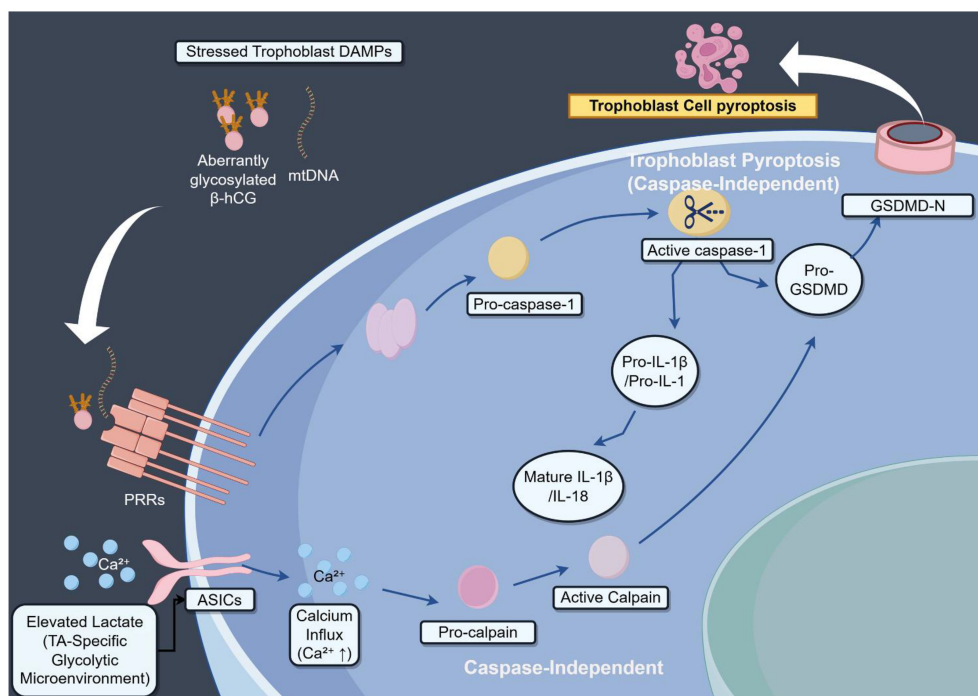


FIGURE 6

Two converging pathways of trophoblast pyroptosis in TA. In the trophoblast-specific glycolytic microenvironment, elevated lactate activates ASICs channels, triggering calcium influx ($\text{Ca}^{2+}\uparrow$). Meanwhile, DAMPs, such as aberrantly glycosylated β -hCG and mtDNA released by stressed trophoblast cells activate PRRs. The calcium influx activates calpain, and PRRs activation induces the maturation of pro-caspase-1 into active caspase-1. Active caspase-1 not only promotes the maturation of pro-IL-1 β /IL-18 into active IL-1 β /IL-18 but also activates pro-GSDMD to form GSDMD-N, ultimately leading to caspase-independent pyroptosis of trophoblast cells and the release of cellular contents. Drawn by FigDraw.

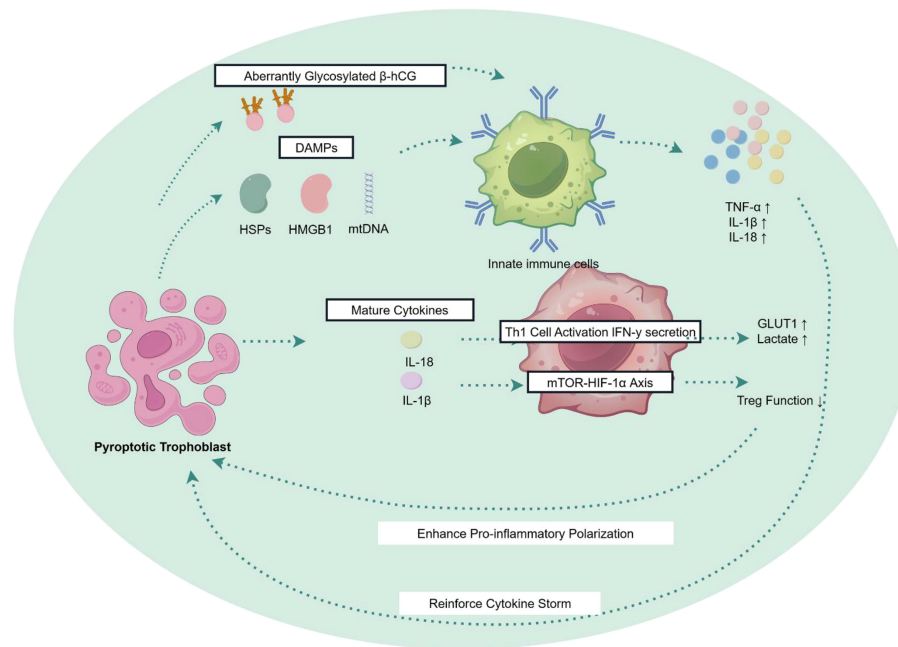


FIGURE 7
Schematic diagram illustrating the vicious cycle of inflammation and tissue damage during the execution stage of TA. Drawn by FigDraw.

immune tolerance. Clinical evidence corroborates these findings: a study involving 44 patients with TA revealed that serum IL-1 β levels were significantly higher in those who experienced miscarriage compared to those with ongoing pregnancies ($P < 0.001$) (66).

Third, aberrantly glycosylated β -hCG released from pyroptotic trophoblasts is present in elevated concentrations, perpetuating the activation of TLR4 and Mincle on innate immune cells (30). This process sustains the initial DAMP signal, thereby ensuring the persistence of the inflammatory response even as trophoblasts undergo cell death. Collectively, this cycle results in extensive placental dysfunction: the pyroptosis of syncytiotrophoblasts compromises the transport of nutrients and oxygen to the fetus, while the pyroptosis of cytotrophoblasts disrupts placental invasion and angiogenesis (67).

5 Translational applications: biomarkers, therapeutics, and challenges

5.1 Stage-specific biomarkers for TA prognosis

The “Danger Signal” Hypothesis facilitates the identification of stage-specific biomarkers that address the limitations inherent in current clinical assessments, such as serum β -hCG and ultrasound, which primarily reflect established dysfunction rather than predictive pathological processes. These biomarkers not only enhance prognostic accuracy but also inform targeted interventions by delineating the pathological stage of TA.

For the initiation stage, the ratio of hCG-h to total hCG serves as a robust predictive marker. A prospective study conducted by Salas et al. demonstrated that the median levels of hCG-h and total hCG in pregnancies with normal outcomes were significantly higher than those in pregnancies resulting in abortion among TA patients (68). Additional glycan-specific markers, such as reduced β -hCG sialylation (quantified by mass spectrometry) and increased α 1, 6-fucosylation (identified through Lens culinaris agglutinin binding), have been associated with trophoblast stress and maternal-fetal immune tolerance (69, 70).

During the amplification stage, biomarkers indicative of metabolic reprogramming exhibit high prognostic value. In a study conducted by Wu et al. involving 60 participants, it was found that Galectin-1 expression levels were significantly lower in patients experiencing spontaneous abortion compared to those undergoing induced abortion during normal early pregnancy (71). Furthermore, flow cytometric analysis of circulating T cells revealed a reduced percentage of CD4+CD25+Foxp3+ regulatory T cells in the spontaneous abortion group compared to the induced abortion group during normal early pregnancy (0.77 ± 0.31 vs. 1.00 ± 0.35 , $P < 0.05$). These results suggest that the downregulation of Galectin-1 and modifications in regulatory T cells may play a pivotal role in the pathogenesis of spontaneous abortion (72). Additionally, lactate metabolism is critical in decidual macrophage-mediated miscarriage, and inhibiting lactate uptake has been shown to ameliorate abortion-prone models (73). Based on the aforementioned studies, the combination of these two markers with β -hCG may serve as biomarkers for TA.

For the execution stage, circulating GSDMD-N fragments (detected via ELISA or Western blot) and serum IL-1 β levels mark irreversible pyroptosis. In the villous tissues of patients with

recurrent spontaneous abortion (RSA), the protein levels of GSDMD-N fragments and IL-1 β are significantly higher than those in the healthy control group (58). A retrospective analysis of 108 patients with threatened abortion has demonstrated that IL-1 β is a risk factor for threatened abortion (74). Similarly, a nested case-control study conducted by Hu et al. revealed that the serum IL-1 β level in the live birth group was significantly higher than that in the abortion group ($P < 0.001$), and serum IL-1 β level was significantly negatively correlated with the risk of abortion (75).

5.2 Targeted therapeutic strategies

The mechanistic understanding of the β -hCG-TNF- α -T cell axis facilitates the development of targeted therapies that address specific stages of TA progression, thereby advancing beyond non-specific interventions such as progesterone supplementation.

In the initial phase, the primary strategy involves restoring normal β -hCG glycosylation by reducing trophoblast stress. Antioxidants, such as N-acetylcysteine (NAC), play a crucial role in mitigating oxidative stress, which is a significant contributor to abnormal glycosylation. A clinical study with 166 women experiencing recurrent miscarriage and elevated oxidative stress markers demonstrated that NAC supplementation (600 mg/day) resulted in a 72% success rate in sustaining pregnancy, compared to a placebo group (76). The effectiveness of NAC intervention was further corroborated in a threatened abortion model (77). Additionally, endoplasmic reticulum stress chaperones, such as 4-phenylbutyrate (4-PBA), enhance glycosylation by restoring glycosyltransferase activity. Research conducted by Liong et al. revealed that 4-phenylbutyric acid alleviates endoplasmic reticulum stress and inflammatory responses, thereby safeguarding gestational tissues and offering direct clinical mechanistic insights for threatened abortion (78).

In the amplification phase, the inhibition of TNF- α and metabolic reprogramming are of paramount importance. Anti-TNF- α agents, including etanercept (a soluble TNFR2 fusion protein), have been shown to mitigate cytokine storms. A randomized controlled trial ($n=188$) demonstrated that the group receiving etanercept (25 mg/week for 4 weeks) exhibited a significant reduction in TNF- α levels and achieved a healthy live birth rate of 89.47%, significantly surpassing the 72.04% observed in the placebo group ($P = 0.01$) (79). In terms of metabolic modulation, metformin, an AMPK activator, counteracts T cell glycolysis by inhibiting mTOR. Treg cells are characterized by a metabolic profile with elevated GLUT1 expression, and metformin can influence cellular metabolism by targeting this characteristic (80).

During the execution stage, inhibiting pyroptosis is crucial to prevent irreversible trophoblast loss. GSDMD inhibitors, such as necrosulfonamide, bind to GSDMD-N to inhibit pore formation. *In vitro* experiments have confirmed that necrosulfonamide (10 μ mol/L) effectively targets the death pathway in trophoblast-related cells (81). For caspase-dependent pyroptosis, NLRP3 inhibitors like MCC950 impede inflammasome activation, thereby reducing IL-1 β secretion and LDH release, and enhancing pregnancy-related outcomes in stressed trophoblast cells (82).

5.3 Unmet challenges in translation

Despite encouraging preclinical and early clinical findings, several challenges must be addressed to facilitate the widespread clinical adoption of these therapies.

Firstly, the safety of these interventions during pregnancy remains a significant concern. Many candidate therapies, such as anti-TNF- α agents and GSDMD inhibitors, lack comprehensive long-term safety data in humans. Beyond inflammatory and cell death inhibitors, metabolic interventions represented by metformin also carry unique safety risks when applied to early threatened abortion. Metformin is widely used for gestational diabetes and polycystic ovary syndrome in pregnancy, but its off-label use for immune metabolic reprogramming in threatened abortion lacks dedicated safety verification. Early pregnancy is a critical window for embryonic organogenesis; metformin-mediated systemic glycolysis inhibition may interfere with physiological energy metabolism of trophoblasts and embryonic cells. A recent primate study further confirmed these concerns. When human-equivalent doses of metformin were given to rhesus monkeys within 30 days of conception, the drug exhibited prominent bioaccumulation in multiple fetal tissues and gestational compartments. The exposed fetuses presented diminished skeletal muscle and retroperitoneal fat, significantly decreased birth weight, and abnormal renal development including disorganized glomerular structure and increased cortical thickness. Notably, these adverse outcomes were not attenuated by maternal dietary patterns (83). Moreover, existing clinical studies mainly focus on patients with pre-gestational metabolic disorders, and there is a paucity of large cohort data clarifying its risk of teratogenicity, adverse fetal weight and long-term metabolic sequelae in fetuses exposed during the first trimester (84). For instance, Etanercept is classified as FDA Category B; however, data regarding fetal exposure and long-term outcomes in infants are limited (85). Future randomized controlled trials (RCTs) should incorporate extended follow-up periods (≥ 2 years) for infants exposed to these interventions to evaluate neurodevelopmental and metabolic outcomes comprehensively.

Secondly, there is a need to improve the accessibility of biomarkers.

Currently, most biomarkers, such as decidual lactate and placental GSDMD-N, require invasive sampling methods (e.g., decidual biopsy) or specialized techniques (e.g., mass spectrometry), which restrict their clinical applicability (86, 87). To overcome limited biomarker accessibility, we herein present a tiered translational framework to translate non-invasive detection assays into clinical use. First, laboratory testing platforms will be converted to point-of-care formats, including lectin-based lateral flow immunoassays (LFAs) for rapid bedside measurement of urinary β -hCG glycoforms such as hCG-h (68) and standardized nanoparticle tracking analysis (NTA) to quantify plasma extracellular vesicle (EV)-derived GSDMD-N and obviate tissue biopsy (88). Second, analytical validation against gold-standard approaches including glycan mass spectrometry and GSDMD-N Western blot will be performed in a 200-subject pilot cohort to establish diagnostic sensitivity, specificity and cut-off thresholds. Last, multicenter

prospective evaluation in >1000 threatened abortion patients with serial baseline, 4-week and 8-week biospecimens will define the biomarkers' predictive performance for pregnancy outcomes, ensuring robust clinical applicability and broad accessibility.

Threatened abortion presents prominent heterogeneity, with patient subgroups modulated by divergent stressors including oxidative stress and intrauterine infection; anti-TNF- α agents exert superior therapeutic effects on infection-related cases, while antioxidants are more suitable for phenotypes induced by oxidative stress (89, 90). Though the proposed β -hCG-TNF- α -T cell axis unifies the pathogenic cascade of threatened abortion, multiple upstream triggers covering oxidative stress, endoplasmic reticulum stress and intrauterine infection converge on placental tissues to elicit the danger signal phenotype with unique kinetic features and biomarker signatures. Infectious stimuli activate pathway amplification through TLR4/MyD88 recognition of pathogen-associated molecular patterns and trigger an early, sharp rise in TNF- α , whereas oxidative stress subtly disturbs glycosyltransferase activity at the pathogenic initiation phase and gradually accumulates abnormal β -hCG glycoforms, highlighting the requirement for stratified biomarker detection. We therefore establish a Dual-Layer Biomarker Strategy: the first layer conducts etiology-targeted screening via *Gardnerella vaginalis* quantification and 8-OHdG measurement to distinguish primary pathogenic insults, and the second layer monitors distinct stages of the core axis using the hCG-h ratio to reflect initiation and TNF- α /IL-1 β to indicate amplification. This two-tiered strategy enables stage- and etiology-matched treatment and avoids improper anti-inflammatory administration when infection clearance is prioritized, and composite panels integrating the hCG-h ratio, IL-1 β concentrations and microbiome profiling should be adopted in subsequent clinical trials to stratify patients and improve therapeutic responsiveness (91).

6 Future perspectives and conclusion

6.1 Priority research directions

To enhance the understanding and management of TA, future research should prioritize three key areas.

Firstly, single-cell and spatial omics approaches will elucidate cell-specific mechanisms (92). Techniques such as single-cell RNA sequencing (scRNA-seq) of decidual tissues will facilitate the identification of T cell subsets, such as Th17 and Treg, that are most susceptible to metabolic reprogramming, as well as trophoblast populations, including syncytiotrophoblasts and extravillous trophoblasts, that are prone to pyroptosis. Spatial transcriptomics is poised to enhance our understanding of the distribution of inflammatory mediators, such as TNF- α and lactate, as well as PRRs like TLR4 and MyD88, within the maternal-fetal interface. This approach is expected to illuminate regional variations in pathological processes, potentially elucidating why certain cases of TA advance to miscarriage while others resolve.

Second, microbiome-immune interactions represent an understudied area (93). The composition of the vaginal and uterine microbiomes plays a critical role in modulating maternal immune tolerance, with

dysbiosis-characterized by an increase in *Gardnerella vaginalis* and a decrease in *Lactobacillus crispatus*-being linked to TA. Future research should focus on elucidating the regulatory effects of microbiome-derived metabolites, such as short-chain fatty acids, on β -hCG glycosylation and trophoblast pyroptosis. Such studies could pave the way for microbiome-based therapeutic interventions, including the use of *Lactobacillus* probiotics, to restore immune tolerance.

Third, longitudinal multi-center studies will validate the "Danger Signal" Hypothesis. These studies should involve prospective cohorts with repeated sampling of serum, urine, and vaginal swabs at intervals of 4–6 weeks to monitor the temporal progression of changes in β -hCG glycosylation, metabolic reprogramming, and pyroptosis in TA (94). These studies will refine biomarker thresholds for each stage and enable the development of a "TA progression score" to guide clinical decision-making.

6.2 Conclusion

This review introduces a comprehensive "Danger Signal" Hypothesis, which delineates the pathogenesis of TA into three sequential and interconnected stages: initiation through aberrant β -hCG glycosylation, amplification via TNF- α -mediated T cell metabolic reprogramming, and execution by trophoblast pyroptosis. By synthesizing insights from glycobiology, immunometabolism, and cell death research, this model transcends mere descriptive associations, offering an explanation for the dynamic transition from reversible trophoblast stress to irreversible pregnancy loss—a significant gap in current TA research.

The translational implications of this framework are significant. Stage-specific biomarkers, such as the hCG-h ratio, GLUT1⁺ T cells, and GSDMD-N, enhance prognostic accuracy, while targeted therapies, including NAC, etanercept, and NSA, address the underlying causes of TA rather than merely alleviating symptoms. Despite ongoing challenges in safety validation, biomarker accessibility, and patient stratification, the proposed hypothesis offers a strategic pathway for the application of precision medicine in the management of TA.

Ultimately, advancing our understanding of the β -hCG-TNF- α -T cell axis is anticipated not only to improve clinical outcomes for TA but also to provide insights into other pregnancy complications, such as preeclampsia, which share similar mechanisms involving inflammation and trophoblast dysfunction. This cross-disease relevance highlights the broader impact of TA research on maternal and fetal health and underscores the necessity for continued investment in mechanistic and translational studies.

Author contributions

SW: Data curation, Writing – original draft. QG: Data curation, Investigation, Writing – original draft. DN: Data curation, Investigation, Writing – original draft. YC: Data curation, Writing – original draft. YP: Data curation, Writing – original draft. JR: Data curation, Writing – original draft. QD: Data curation, Writing – original draft. XF: Formal Analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This work was supported by the Traditional Chinese Medicine Innovation Team and Talent Support Program Project (Grant No. ZYYCXTD-C-202404) and the National Natural Science Foundation of China (Grant No. 82575134).

Acknowledgments

We would appreciate the participation of each participant.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

- Hendriks E, MacNaughton H, MacKenzie MC. First trimester bleeding: evaluation and management. *Am Fam Physician*. (2019) 99(3):166–174.
- Yang M, Luo J, Yang Q, Xu L. Research on the medication rules of Chinese herbal formulas on treatment of threatened abortion. *Complement Ther Clin Pract*. (2021) 43:101371. doi: 10.1016/j.ctcp.2021.101371
- Quilon M, Glucksman A, Emmanuel G, Greenstein J, Hahn B. Female with vaginal bleeding. *Clin Pract Cases Emerg Med*. (2020) 4(4):636–637. doi: 10.5811/cpcem.2020.8.48627
- McLindon LA, James G, Beckmann MM, Bertolone J, Mahomed K, Vane M, et al. Progesterone for women with threatened miscarriage (STOP trial): a placebo-controlled randomized clinical trial. *Hum Reprod*. (2023) 38(4):560–568. doi: 10.1093/humrep/dead029
- Zhu Z, Wei N, Guo J, Yue C, Chen C, Zhang Z, et al. Predicting the risk of threatened abortion using machine learning methods: a comparative study. *BMC Pregnancy Childbirth*. (2025) 25(1):901. doi: 10.1186/s12884-025-08030-z
- Hinney B, Henze C, Kuhn W, Wuttke W. The corpus luteum insufficiency: a multifactorial disease. *J Clin Endocrinol Metab*. (1996) 81(2):565–570. doi: 10.1210/jcem.81.2.8636265
- Huchon C, Deffieux X, Beucher G, Capmas P, Carcopino X, Costedoat-Chalumeau N, et al. Pregnancy loss: French clinical practice guidelines. *Eur J Obstet Gynecol Reprod Biol*. (2016) 201:18–26. doi: 10.1016/j.ejogrb.2016.02.015
- Marsters CM, Staff L, Bugden S, Gustainis R, Nkunu V, Reimer R, et al. Pregnancy, obstetrical and neonatal outcomes in women exposed to physician-related occupational hazards: a scoping review. *BMJ Open*. (2023) 13(2):e064483. doi: 10.1136/bmjopen-2022-064483
- Paradisi R, Porcu E, Venturoli S, Maldini-Casadei M, Boni P. Maternal serum levels of pro-inflammatory cytokines in missed and threatened abortion. *Am J Reprod Immunol*. (2003) 50(4):302–308. doi: 10.1034/j.1600-0897.2003.00065.x
- Turgut E, Yildirim M, Sakcak B, Ayhan SG, Tekin OM, Sahin D. Predicting miscarriage using systemic immune-inflammation index. *J Obstet Gynaecol Res*. (2022) 48(3):587–592. doi: 10.1111/jog.15156
- Zhao Y, Miao C, Wang R, Chen Y, Ren N, Ma J, et al. Jianpi Antai formula prevents miscarriage by repressing M1 polarization of decidual macrophages through ubiquitination of NLRP3 mediated by MARCH7. *J Ethnopharmacol*. (2024) 324:117796. doi: 10.1016/j.jep.2024.117796
- Gucer F, Balkanli-Kaplan P, Yuksel M, Sayin NC, Yuce MA, Yardim T. Maternal serum levels of tumor necrosis factor-alpha and interleukin-2 receptor in threatened abortion: a comparison with normal and pathologic pregnancies. *Fertil Steril*. (2001) 76(4):707–711. doi: 10.1016/s0015-0282(01)02002-7
- Herkert D, Meljen V, Muasher L, Price TM, Kuller JA, Dotters-Katz S. Human chorionic gonadotropin-a review of the literature. *Obstet Gynecol Surv*. (2022) 77(9):539–546. doi: 10.1097/ogx.0000000000001053

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

- Feng W, Matzuk MM, Mountjoy K, Bedows E, Ruddon RW, Boime I. The asparagine-linked oligosaccharides of the human chorionic gonadotropin beta subunit facilitate correct disulfide bond pairing. *J Biol Chem*. (1995) 270(20):11851–11859. doi: 10.1074/jbc.270.20.11851
- Cole LA. hCG, the wonder of today's science. *Reprod Biol Endocrinol*. (2012) 10:24. doi: 10.1186/1477-7827-10-24
- Wu Z, Zhang L, Jia Y, Bi B, Fang L, Cheng JC. GDF-11 downregulates placental human chorionic gonadotropin expression by activating SMAD2/3 signaling. *Cell Commun Signal*. (2023) 21(1):179. doi: 10.1186/s12964-023-01201-5
- Polese B, Gridelet V, Arakioti E, Martens H, Perrier d'Hauterive S, Geenen V. The endocrine milieu and CD4 T-lymphocyte polarization during pregnancy. *Front Endocrinol (Lausanne)*. (2014) 5:106. doi: 10.3389/fendo.2014.00106
- Nakamura K, Niimi K, Yamamoto E, Ikeda Y, Nishino K, Suzuki S, et al. Core 2 beta1,6-N-acetylglucosaminyltransferases accelerate the escape of choriocarcinoma from natural killer cell immunity. *Biochem Biophys Res*. (2021) 26:100951. doi: 10.1016/j.bbrep.2021.100951
- Jing G, Yao J, Dang Y, Liang W, Xie L, Chen J, et al. The role of beta-HCG and VEGF-MEK/ERK signaling pathway in villi angiogenesis in patients with missed abortion. *Placenta*. (2021) 103:16–23. doi: 10.1016/j.placenta.2020.10.005
- Vornic I, Buciu V, Fura CG, Gaje PN, Ceausu RA, Dumitru CS, et al. Oxidative stress and placental pathogenesis: a contemporary overview of potential biomarkers and emerging therapeutics. *Int J Mol Sci*. (2024) 25(22):12195. doi: 10.3390/ijms252212195
- Yoshida K, Kusama K, Fukushima Y, Ohmaru-Nakanishi T, Kato K, Tamura K. Alpha-1 antitrypsin-induced endoplasmic reticulum stress promotes invasion by extravillous trophoblasts. *Int J Mol Sci*. (2021) 22(7):3683. doi: 10.3390/ijms22073683
- McDonald AG, Hayes JM, Bezak T, Gluchowska SA, Cosgrave EF, Struwe WB, et al. Galactosyltransferase 4 is a major control point for glycan branching in N-linked glycosylation. *J Cell Sci*. (2014) 127(23):5014–5026. doi: 10.1242/jcs.151878
- Kobata A, Takeuchi M. Structure, pathology and function of the N-linked sugar chains of human chorionic gonadotropin. *Biochim Biophys Acta*. (1999) 1455(2-3):315–326. doi: 10.1016/s0925-4439(99)00060-5
- Einerson BD, Straubhaar A, Soisson S, Szcotka K, Dodson MK, Silver RM, et al. Hyperglycosylated hCG and placenta accreta spectrum. *Am J Perinatol*. (2019) 36(1):22–26. doi: 10.1055/s-0038-1636501
- Sasaki Y, Ladner DG, Cole LA. Hyperglycosylated human chorionic gonadotropin and the source of pregnancy failures. *Fertil Steril*. (2008) 89(6):1781–1786. doi: 10.1016/j.fertnstert.2007.03.010
- Parvanov D, Nikolova D, Ganeva R, Nikolova K, Vasileva M, Rangelov I, et al. Unbalanced human embryos secrete more hyperglycosylated human chorionic gonadotropin (hCG-H) than balanced ones. *J Assist Reprod Genet*. (2020) 37(6):1341–1348. doi: 10.1007/s10815-020-01776-9

27. Herghelegiu CG, Veduta A, Stefan MF, Magda SL, Ionascu I, Radoi VE, et al. Hyperglycosylated-hCG: its role in trophoblast invasion and intrauterine growth restriction. *Cells*. (2023) 12(12):1647. doi: 10.3390/cells12121647
28. Evans J. Hyperglycosylated hCG: a unique human implantation and invasion factor. *Am J Reprod Immunol*. (2016) 75(3):333–340. doi: 10.1111/aji.12459
29. Phelps J, Leonard C, Shah S, Krawetz R, Hart DA, Duncan NA, et al. Production of mesenchymal progenitor cell-derived extracellular vesicles in suspension bioreactors for use in articular cartilage repair. *Stem Cells Transl Med*. (2022) 11(1):73–87. doi: 10.1093/stcltm/szab008
30. Cao X, Zhou M, Wang C, Hou L, Zeng B. Lectin purified from *Musca domestica* pupa up-regulates NO and iNOS production via TLR4/NF-kappaB signaling pathway in macrophages. *Int Immunopharmacol*. (2011) 11(4):399–405. doi: 10.1016/j.intimp.2010.12.004
31. Nagata M, Izumi Y, Ishikawa E, Kiyotake R, Doi R, Iwai S, et al. Intracellular metabolite beta-glucosylceramide is an endogenous Mincle ligand possessing immunostimulatory activity. *Proc Natl Acad Sci USA*. (2017) 114(16):E3285–E3294. doi: 10.1073/pnas.1618133114
32. Garcia-Garcia A, Ceballos-Laita L, Serna S, Artschwager R, Reichardt NC, Corzana F, et al. Structural basis for substrate specificity and catalysis of alpha1,6-fucosyltransferase. *Nat Commun*. (2020) 11(1):973. doi: 10.1038/s41467-020-14794-z
33. Gringhuis SI, Kaptein TM, Wevers BA, van der Vlist M, Klaver EJ, van Die I, et al. Fucose-based PAMPs prime dendritic cells for follicular T helper cell polarization via DC-SIGN-dependent IL-27 production. *Nat Commun*. (2014) 5:5074. doi: 10.1038/ncomms6074
34. Kharfi A, Leblanc S, Ouellet A, Moutquin JM. Dual action of H2O2 on placental hCG secretion: implications for oxidative stress in preeclampsia. *Clin Biochem*. (2007) 40(1–2):94–97. doi: 10.1016/j.clinbiochem.2006.10.008
35. Kharfi A, Giguere Y, De Grandpre P, Moutquin JM, Forest JC. Human chorionic gonadotropin (hCG) may be a marker of systemic oxidative stress in normotensive and preeclamptic term pregnancies. *Clin Biochem*. (2005) 38(8):717–721. doi: 10.1016/j.clinbiochem.2005.04.011
36. Fu B, Wei H. Decidual natural killer cells and the immune microenvironment at the maternal-fetal interface. *Sci China Life Sci*. (2016) 59(12):1224–1231. doi: 10.1007/s11427-016-0337-1
37. Ali Rahmani S, Paknejad Z, Mohammadkhanlou M. The effects of tumor necrosis factor-alpha (TNF-alpha) and IL-1 receptor antagonist (IL-1Ra) polymorphisms on recurrent abortion in Azari women. *Ginek Pol*. (2017) 88(8):421–427.
38. Piosik ZM, Goegebeur Y, Klitkou L, Steffensen R, Christiansen OB. Plasma TNF-alpha levels are higher in early pregnancy in patients with secondary compared with primary recurrent miscarriage. *Am J Reprod Immunol*. (2013) 70(5):347–358. doi: 10.1111/aji.12135
39. Zhang C, Deng X, Zhang X, Pan Z, Zhao W, Zhang Y, et al. Association between serum TNF-alpha levels and recurrent spontaneous miscarriage: a meta-analysis. *Am J Reprod Immunol*. (2016) 75(2):86–93. doi: 10.1111/aji.12447
40. Ali A, Biswas A, Pal M. HSF1 mediated TNF-alpha production during proteotoxic stress response pioneers proinflammatory signal in human cells. *FASEB J*. (2019) 33(2):2621–2635. doi: 10.1096/fj.201801482r
41. Parkash J. Tumor necrosis factor-alpha induces transcriptional activation of nuclear factor-kappaB in insulin-producing beta-cells. *Exp Ther Med*. (2011) 2(1):21–26. doi: 10.3892/etm.2010.161
42. Xu ZP, Devillier P, Xu GN, Qi H, Zhu L, Zhou W, et al. TNF-alpha-induced CXCL8 production by A549 cells: involvement of the non-neuronal cholinergic system. *Pharmacol Res*. (2013) 68(1):16–23. doi: 10.1016/j.phrs.2012.10.016
43. Renaud SJ, Sullivan R, Graham CH. Tumour necrosis factor alpha stimulates the production of monocyte chemoattractants by extravillous trophoblast cells via differential activation of MAPK pathways. *Placenta*. (2009) 30(4):313–319. doi: 10.1016/j.placenta.2009.01.001
44. Elazab R, Alkhiary M, Bedairi M, Wageh A. Simultaneous use of tumor necrosis factor, lipid profile, and beta-hCG as markers of severity of preeclampsia. *J Obstet Gynaecol India*. (2022) 72(Suppl 1):83–88. doi: 10.1007/s13224-021-01556-0
45. Wang R, Chen L, Wang X, Liu Y. Association between serum beta-human chorionic gonadotropin and inflammation, oxidative stress in pregnancy-induced hypertension. *Microvasc Res*. (2021) 135:104130. doi: 10.1016/j.mvr.2020.104130
46. Li X, Wu PL, Zhu JW, Xue Q, Yang HX. Effect and mechanism of TNF-alpha and etanercept on the invasion ability of extravillous trophoblast cell in URSA patients. *Zhonghua Fu Chan Ke Za Zhi*. (2021) 56(10):705–711. doi: 10.3760/cma.j.cn112141-20210610-00315
47. Shi LZ, Wang R, Huang G, Vogel P, Neale G, Green DR, et al. HIF1alpha-dependent glycolytic pathway orchestrates a metabolic checkpoint for the differentiation of TH17 and Treg cells. *J Exp Med*. (2011) 208(7):1367–1376. doi: 10.1084/jem.20110278
48. Zhao X, Zhang J, Li C, Kuang W, Deng J, Tan X, et al. Mitochondrial mechanisms in Treg cell regulation: implications for immunotherapy and disease treatment. *Mitochondrion*. (2025) 80:101975. doi: 10.1016/j.mito.2024.101975
49. Remels AH, Gosker HR, Verhees KJ, Langen RC, Schols AM. TNF-alpha-induced NF-kappaB activation stimulates skeletal muscle glycolytic metabolism through activation of HIF-1alpha. *Endocrinology*. (2015) 156(5):1770–1781. doi: 10.1210/en.2014-1916
50. Bishop EL, Gudgeon N, Fulton-Ward T, Stavrou V, Roberts J, Boufersaoui A, et al. TNF-alpha signals through ITK-Akt-mTOR to drive CD4(+) T cell metabolic reprogramming, which is dysregulated in rheumatoid arthritis. *Sci Signal*. (2024) 17(833):eadg5678. doi: 10.1126/scisignal.adg5678
51. Haddad JJ, Land SC. A non-hypoxic, ROS-sensitive pathway mediates TNF-alpha-dependent regulation of HIF-1alpha. *FEBS Lett*. (2001) 505(2):269–274. doi: 10.1016/s0014-5793(01)02833-2
52. Tsun A, Chen Z, Li B. Romance of the three kingdoms: RORgammaT allies with HIF1alpha against FoxP3 in regulating T cell metabolism and differentiation. *Protein Cell*. (2011) 2(10):778–781. doi: 10.1007/s12338-011-1114-2
53. Suematsu N, Tsutsui H, Wen J, Kang D, Ikeuchi M, Ide T, et al. Oxidative stress mediates tumor necrosis factor-alpha-induced mitochondrial DNA damage and dysfunction in cardiac myocytes. *Circulation*. (2003) 107(10):1418–1423. doi: 10.1161/01.cir.0000055318.09997.1f
54. Stadler J, Bentz BG, Harbrecht BG, Di Silvio M, Curran RD, Billiar TR, et al. Tumor necrosis factor alpha inhibits hepatocyte mitochondrial respiration. *Ann Surg*. (1992) 216(5):539–546. doi: 10.1097/0000658-199211000-00003
55. Huppertz B, Kingdom JC. Apoptosis in the trophoblast—role of apoptosis in placental morphogenesis. *J Soc Gynecol Investig*. (2004) 11(6):353–362. doi: 10.1016/j.jsg.2004.06.002
56. Weir A, Vince JE. No longer married to inflammasome signaling: the diverse interacting pathways leading to pyroptotic cell death. *Biochem J*. (2022) 479(10):1083–1102. doi: 10.1042/bj20210711
57. Shi Y, Hu Y, Li J, Chen H, Zhong Q, Ma X, et al. Inhibition of caspase-1 suppresses GSDMD-mediated peritoneal mesothelial cell pyroptosis and inflammation in peritoneal fibrosis. *Small*. (2025) 21(14):e2409362. doi: 10.1002/smll.202409362
58. Wang R, Xu X, Yang J, Chen W, Zhao J, Wang M, et al. BPDE exposure promotes trophoblast cell pyroptosis and induces miscarriage by up-regulating lnc-HZ14/ZBP1/NLRP3 axis. *J Hazard Mater*. (2023) 455:131543. doi: 10.1016/j.jhazmat.2023.131543
59. Xu M, Ji X, Zhang H, Zeng D, Zhao Z, Liu J, et al. Anti-beta2GPI antibodies induce trophoblast pyroptosis and placental inflammation through TLR4/NLRP3/GSDMD axis. *Placenta*. (2025) 170:64–74. doi: 10.1016/j.placenta.2025.08.330
60. Ma LN, Huang XB, Muyayalo KP, Mor G, Liao AH. Lactic acid: a novel signaling molecule in early pregnancy?. *Front Immunol*. (2020) 11:279. doi: 10.3389/fimmu.2020.00279
61. Li X, Wu FR, Xu RS, Hu W, Jiang DL, Ji C, et al. Acid-sensing ion channel 1a-mediated calcium influx regulates apoptosis of endplate chondrocytes in intervertebral discs. *Expert Opin Ther Targets*. (2014) 18(1):1–14. doi: 10.1517/14728222.2014.859248
62. Cheng SB, Nakashima A, Huber WJ, Davis S, Banerjee S, Huang Z, et al. Pyroptosis is a critical inflammatory pathway in the placenta from early onset preeclampsia and in human trophoblasts exposed to hypoxia and endoplasmic reticulum stressors. *Cell Death Dis*. (2019) 10(12):927. doi: 10.1038/s41419-019-2162-4
63. Maisat W, Yuki K. The role of damage-associated molecular patterns in perioperative neurocognitive disorders: a narrative review. *Anesth Analg*. (2026) 142(2):312–325. doi: 10.1213/ane.0000000000007682
64. Huang F, Wu J, Zheng Q. HMGB1 as a trigger for inflammatory storms: a potential biomarker for adverse pregnancy outcomes. *Biol Reprod*. (2026) 114(3):687–699. doi: 10.1093/biolre/iaof243
65. Hellwig-Burgel T, Rutkowski K, Metzner E, Fandrey J, Jelkmann W. Interleukin-1beta and tumor necrosis factor-alpha stimulate DNA binding of hypoxia-inducible factor-1. *Blood*. (1999) 94(5):1561–1567. doi: 10.1182/blood.v94.5.1561
66. Vitoratos N, Papadias C, Economou E, Makrakis E, Panoulis C, Creatsas G. Elevated circulating IL-1beta and TNF-alpha, and unaltered IL-6 in first-trimester pregnancies complicated by threatened abortion with an adverse outcome. *Mediators Inflamm*. (2006) 2006(4):30485. doi: 10.1155/mi/2006/30485
67. Sharp AN, Heazell AE, Crocker IP, Mor G. Placental apoptosis in health and disease. *Am J Reprod Immunol*. (2010) 64(3):159–169. doi: 10.1111/j.1600-0897.2010.00837.x
68. Salas A, Gaston B, Barrenetxea J, Sendino T, Jurado M, Alcazar JL. Predictive value of hyperglycosylated human chorionic gonadotropin for pregnancy outcomes in threatened abortion in first-trimester viable pregnancies. *An Sist Sanit Navar*. (2021) 44(1):23–31. doi: 10.23938/ASSN.0933
69. Orczyk-Pawilowicz M, Augustyniak D, Hirnle L, Katnik-Prastowska I. Degree of sialylation and fucosylation of plasma and amniotic immunoglobulin G changes progressively during normal pregnancy. *Prenat Diagn*. (2012) 32(5):432–439. doi: 10.1002/pd.3832
70. Elliott MM, Kardana A, Lustbader JW, Cole LA. Carbohydrate and peptide structure of the alpha- and beta-subunits of human chorionic gonadotropin from normal and aberrant pregnancy and choriocarcinoma. *Endocrine*. (1997) 7(1):15–32. doi: 10.1007/bf02778058

71. Wu M, Liu P, Cheng L. Galectin-1 reduction and changes in T regulatory cells may play crucial roles in patients with unexplained recurrent spontaneous abortion. *Int J Clin Exp Pathol.* (2015) 8(2):1973–1978.
72. Ramhorst RE, Giribaldi L, Fraccaroli L, Toscano MA, Stupirski JC, Romero MD, et al. Galectin-1 confers immune privilege to human trophoblast: implications in recurrent fetal loss. *Glycobiology.* (2012) 22(10):1374–1386. doi: 10.1093/glycob/cws104
73. Gao L, Xu QH, Ma LN, Luo J, Muyayalo KP, Wang LL, et al. Trophoblast-derived lactic acid orchestrates decidual macrophage differentiation via SRC/LDHA signaling in early pregnancy. *Int J Biol Sci.* (2022) 18(2):599–616. doi: 10.7150/ijbs.67816
74. Yang S, Wang M, Yang L, Lin N. Diagnostic value of color Doppler ultrasound combined with blood inflammatory markers in threatened abortion and pregnancy outcomes in early pregnancy. *Acta Radiol.* (2025) 66(7):733–739. doi: 10.1177/02841851251324923
75. Hu Y, Zhang D, Zhang Q, Yin T, Jiang T, He S, et al. Serum Cu, Zn and IL-1beta levels may predict fetal miscarriage risk after IVF cycles: a nested case-control study. *Biol Trace Elem Res.* (2023) 201(12):5561–5574. doi: 10.1007/s12011-023-03621-3
76. Amin AF, Shaaban OM, Bediawy MA. N-acetyl cysteine for treatment of recurrent unexplained pregnancy loss. *Reprod Biomed Online.* (2008) 17(5):722–726. doi: 10.1016/s1472-6483(10)60322-7
77. Liu Y, Na Q, Liu J, Liu A, Oppong A, Lee JY, et al. Dendrimer-based N-acetyl cysteine maternal therapy ameliorates placental inflammation via maintenance of M1/M2 macrophage recruitment. *Front Bioeng Biotechnol.* (2022) 10:819593. doi: 10.3389/fbioe.2022.819593
78. Liong S, Lappas M. Endoplasmic reticulum stress is increased after spontaneous labor in human fetal membranes and myometrium where it regulates the expression of prolabor mediators. *Biol Reprod.* (2014) 91(3):70. doi: 10.1095/biolreprod.114.120741
79. Fu J, Li L, Qi L, Zhao L. A randomized controlled trial of etanercept in the treatment of refractory recurrent spontaneous abortion with innate immune disorders. *Taiwan J Obstet Gynecol.* (2019) 58(5):621–625. doi: 10.1016/j.tjog.2019.07.007
80. Kunisada Y, Eikawa S, Tomonobu N, Domae S, Uehara T, Hori S, et al. Attenuation of CD4(+)CD25(+) regulatory T cells in the tumor microenvironment by metformin, a type 2 diabetes drug. *EBioMedicine.* (2017) 25:154–164. doi: 10.1016/j.ebiom.2017.10.009
81. Qin C, Wu J, Wei X, Liu X, Lin Y. ALKBH5 modulation of ferroptosis in recurrent miscarriage: implications in cytotrophoblast dysfunction. *PeerJ.* (2024) 12:e18227. doi: 10.7717/peerj.18227
82. Gomez-Lopez N, Romero R, Garcia-Flores V, Leng Y, Miller D, Hassan SS, et al. Inhibition of the NLRP3 inflammasome can prevent sterile intra-amniotic inflammation, preterm labor/birth, and adverse neonatal outcomes. *Biol Reprod.* (2019) 100(5):1306–1318. doi: 10.1093/biolre/iy0264
83. Bolte E, Dean T, Garcia B, Seferovic MD, Sauter K, Hummel G, et al. Initiation of metformin in early pregnancy results in fetal bioaccumulation, growth restriction, and renal dysmorphology in a primate model. *Am J Obstet Gynecol.* (2024) 231(3):352.e1–352.e16. doi: 10.1016/j.ajog.2024.06.002
84. Paschou SA, Shalit A, Gerontiti E, Athanasiadou KI, Kalampokas T, Psaltopoulou T, et al. Efficacy and safety of metformin during pregnancy: an update. *Endocrine.* (2024) 83(2):259–269. doi: 10.1007/s12020-023-03550-0
85. Carman WJ, Accortt NA, Anthony MS, Iles J, Enger C. Pregnancy and infant outcomes including major congenital malformations among women with chronic inflammatory arthritis or psoriasis, with and without etanercept use. *Pharmacoepidemiol Drug Saf.* (2017) 26(9):1109–1118. doi: 10.1002/pds.4261
86. Wheeler KC, Jena MK, Pradhan BS, Nayak N, Das S, Hsu CD, et al. VEGF may contribute to macrophage recruitment and M2 polarization in the decidua. *PLoS One.* (2018) 13(1):e0191040. doi: 10.1371/journal.pone.0191040
87. Wang LL, Liu H, Zhao SJ, Shen L, Xie T, Luo J, et al. The metabolic landscape of decidua in recurrent pregnancy loss using a global metabolomics approach. *Placenta.* (2021) 112:45–53. doi: 10.1016/j.placenta.2021.07.001
88. Kowkabany G, Bao Y. Nanoparticle tracking analysis: an effective tool to characterize extracellular vesicles. *Molecules.* (2024) 29(19):4672. doi: 10.3390/molecules29194672
89. Rychly DJ, DiPiro JT. Infections associated with tumor necrosis factor-alpha antagonists. *Pharmacotherapy.* (2005) 25(9):1181–1192. doi: 10.1592/phco.2005.25.9.1181
90. Sakai T, Imai J, Ito T, Takagaki H, Ui M, Hatta S. The novel antioxidant TA293 reveals the role of cytoplasmic hydroxyl radicals in oxidative stress-induced senescence and inflammation. *Biochem Biophys Res Commun.* (2017) 482(4):1183–1189. doi: 10.1016/j.bbrc.2016.12.009
91. Chu HN, Zhou FZ, Xie X. Relationship between imbalanced synthesis of human chorionic gonadotropin alpha and beta subunits and pregnancy-induced hypertension. *Zhejiang Da Xue Xue Bao Yi Xue Ban.* (2004) 33(6):529–531. doi: 10.3785/j.issn.1008-9292.2004.06.013
92. Gulati GS, D'Silva JP, Liu Y, Wang L, Newman AM. Profiling cell identity and tissue architecture with single-cell and spatial transcriptomics. *Nat Rev Mol Cell Biol.* (2025) 26(1):11–31. doi: 10.1038/s41580-024-00768-2
93. Lin NY, Fukuoka S, Koyama S, Motooka D, Tourlousse DM, Shigeno Y, et al. Microbiota-driven antitumour immunity mediated by dendritic cell migration. *Nature.* (2025) 644(8078):1058–1068. doi: 10.1038/s41586-025-09249-8
94. Vasu S, Zhao Q, Miller EG, Elder P, Langenberg L, Cataland S, et al. High incidence of severe TA-TMA increases mortality in adult allogeneic transplant recipients: a prospective MIDAS Consortium study. *Blood.* (2025) 146(5):638–646. doi: 10.1182/blood.2025028390